stellar systems flung off into space. Over time, though, it will become increasingly and terrifyingly clear that something very different is going on.

As galaxies get closer together and merge more frequently, galaxies across the sky will burst with the blue light of new stars, and giant jets of particles and radiation will rip through the intergalactic gas. New planets might be born along with those new stars, and perhaps some will have time to develop life, though the terrifying frequency of supernovae in this chaotic, collapsing universe might irradiate the new planets clean. The violence of the gravitational interactions between galaxies and between central supermassive black holes will increase, flinging stars out of their own galaxies to end up caught in the gravity of others. But even at this point, collisions of individual stars will be rare, and they will remain so until very late in the game. The destruction of stars comes about through another process, one that also ensures, with great finality, the destruction of any planetary life that might still be lingering on.

Here's how.

The expansion of the universe as it is occurring today does more than just stretch out the light of distant galaxies. It also stretches out and dilutes the afterglow of the Big Bang itself. One of the strongest pieces of evidence for the Big Bang, discussed in the previous chapter, is the fact that we can *actually see it*, simply by looking far enough away. What we see, specifically, is a dim glow, coming from all directions, of light produced in the universe's infancy. That dim glow is actually a direct view of parts of the universe that are so far away that, from our perspective, they are still *on fire*—they're still experiencing the hot early stage of the universe's existence, when every part of the cosmos was hot and dense and opaque with roiling plasma, like the inside of a star. The light from that long-burned-out fire has been traveling to us all this time, and, from sufficiently distant points, has just now arrived.

The reason we experience this as a low-energy, diffuse background (the cosmic microwave background) is that the expansion of the universe has stretched out and separated the individual photons to the point that they're now merely a bit of faint static. And the fact that they show up as microwaves is due to extreme redshifting. The expansion of the universe can do a lot, including taking the heat of an unimaginable inferno and diluting and stretching it out until it's just a faint microwave hum we might experience only as a tiny bit of static on an old-fashioned analog TV.